

JIN HUANG

Applied Researcher, Generative AI

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EDUCATION

Ph.D., Computer Science and Engineering – University of Notre Dame, IN, USA (2025)

Master of Science, Computer Science – New York University, NY, USA (2018)

Bachelor of Engineering, Software Engineering – Nankai University, Tianjin, China (2016)

RESEARCH DIRECTIONS AND SKILLS

Research Fields: Computer Vision, Image/Video Generation and Editing, VLM, Reinforcement Learning, Model Evaluation

Skills: Python, PyTorch, Hugging Face, Diffusion Models, Transformers, Supervised Fine-Tuning (SFT), Parameter-Efficient Fine-Tuning (PEFT/LoRA), Synthetic Data Generation, Data Curation, Prompt Engineering.

WORK AND RESEARCH EXPERIENCES

Applied Researcher, Generative AI

Meitu Inc. – Pixocial Technology (Subsidiary), MT-Lab R&D Department. Bellevue, WA, USA. Jun 2025 – Now

- Developed a novel **controllable image-editing diffusion model** that enables **continuous transformations** between source and target images through **adjustable editing strength** for specific visual attributes.
- Designed and built two end-to-end large scale **synthetic data pipelines** using image and video generation models to produce transformation sequences; orchestrated open- and closed-source models through iterative **prompt engineering** and human supervision, and developed a dedicated downstream **VLM filtering pipeline**, yielding over 100K sequences across 80 editing attributes for continuous image editing.
- Conducted large-scale **post-training** and **evaluation** across supervision strategies, data sources, and editing attributes, improving transformation **continuity and identity preservation** and demonstrating **controllable editing** through high quality demos on multiple visual attributes.
- Led end-to-end development of a novel Wan 2.2-based, **mask-free video-editing model** inspired by FLUX.1 Kontext, re-designing the architecture to enable **global VFX transformations** with scene-structure preservation.
- Designed and implemented synthetic **data generation, filtering, training, and evaluation** pipelines using Runway-generated video pairs; demonstrated **weather-editing capabilities** in selected cases and supported engineering-led product demos.
- Mentored several interns on research topics across **image and video generation and editing, video understanding, VLM, model evaluation and reinforcement learning**, guiding research direction exploration, data collection, experimental design and manuscript writing and published in top-tier venues.

Deep Learning Intern

W. W. Grainger, Grainger Technology Group (GTG). Lake Forest, IL, USA. Jun 2024 – Aug 2024

- Constructed pipeline for data cleaning and analysis on over 100 categories of Grainger industrial data.
- Improved retrieval accuracy and recall by fine-tuning Multi-Modal Foundational Models and Large Language Models and utilizing extracted multi-modal features.

Computer Vision Research Intern

Dolby Laboratories, Advanced Technology Group (ATG). Sunnyvale, CA, USA. Jun 2023 – Sep 2023

- Completed data collection and analysis on human gaze data for real and AI-Generated face recognition. First author paper accepted to Automatic Face and Gesture Recognition (FG) 2025 as an oral presentation.
- Explored human perception inspired method for improving StylGAN-3 for better image generation.
- Co-invented "Enhancement of Generative Image Models Based on Gaze" system, resulting in a filed US patent.

Computer Vision R&D Engineer Intern

Kitware Inc, Computer Vision Team. Clifton Park, NY, USA. May 2021 - Aug 2021.

- Improved data pipeline, model training and testing on Human Activity Recognition (HAR) and Open-World Learning (OWL).
 - Explored Contrastive Learning, Representation Learning and Self-Supervised Learning for HAR.
 - Co-authored a paper that was later accepted to Journal of Artificial Intelligence (JAIR).
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Graduate Research Assistant

Notre Dame Computer Vision Research Lab (CVRL). Notre Dame, IN, USA. Jan 2020 – Jun 2025

- Developed an Extreme Value Theory-based method for modeling category decision boundaries in visual recognition.
 - Applied multi-objective optimization to novel grass segmentation and ecological monitoring.
 - Formalized and benchmarked human activity recognition in an Open World setting.
 - Developed a human-perception-guided deep learning method for Open-Set Recognition.
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Computer Vision Research Engineer

Singularity.AI. Bellevue, WA, USA. Jun 2018 – May 2019

- Developed and trained deep learning based models for frozen section tumor segmentation and classification.
 - Explored multi-Modal representation learning and image enhancement on CT and MRI images.
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Graduate Research Assistant

NYU Multi Medial and Visual Computing Lab (MMVC). New York, NY, USA. Mar 2017 – May 2018

- Implmented Deep Learning based text localization and recognition methods in complex nature scenes.
- Explored multi-organ segmentation in CT and MRI images with deep adversarial networks (Master thesis).

SELECTED PUBLICATIONS

- Tan, Y., Huang, J., **Jin Huang**, Li, H., & Gong, B. (2026). *Natural language camera movement understanding*. The European Conference on Computer Vision (ECCV) 2026.
- **Jin Huang**, Arun, D., & Scheirer, W. (2025). Extreme value theory for modeling category decision boundaries in visual recognition. biorxiv: <https://www.biorxiv.org/content/10.1101/2025.09.08.673121v1>.
- Abraham, S., **Jin Huang**, RichardWebster, B., Milford, M., & Scheirer., W. (2024). Enhancing ecological monitoring with multi-objective optimization: A novel dataset and methodology for segmentation algorithms. arxiv: 2408.06356.
- **Jin Huang**, Gopalakrishnan, S., Mittal, T., Zuen, J., & Pytlarz, J. (2024). *A human eye tracking study on real and fake face recognition based on generative models*. International Conference on Automatic Face and Gesture Recognition (FG 2025).
- Gopalakrishnan, S., Mittal, T., **Jin Huang**, Zuen, J., & Pytlarz, J. (2024). *Enhancement of generative image models based on gaze*. US Patent No. 63/553,406. Filed by Dolby Laboratories in Feb 2024.
- Prijatelj, D., Grieggs, S., **Huang, Jin**, & Scheirer, W. (2024). *Human activity recognition in an open world*. Journal of Artificial Intelligence Research (JAIR) 81 (2024) 85-122. Arxiv: <https://arxiv.org/abs/2212.12141>.
- **Jin Huang**, Prijatelj, D., Duley, J., & Scheirer, W. (2023). *Measuring human perception to improve open-set recognition*. IEEE Transaction on Pattern Analysis and Machine Intelligence (T-PAMI). <https://arxiv.org/abs/2209.03519>.

MISCELLANEOUS

- Reviewer: T-IFS, T-PAMI, CVIU, CVPR, ECCV, ICCV, WACV, FG, BMVC.
- Conference talk: "Extreme Value Theory for Modeling Category Decision Boundaries in Visual Recognition." Vision Science Society (VSS) Annual Meeting, May 2025, Florida.
- Invited talk: "Measuring Human Perception to Improve Open Set Recognition." 2nd MSU-ND Computer Vision and Biometrics Workshop, May 2023, Michigan State University.
- Best Presentation Award for Doctoral Consortium: "Visual Psychophysics for Open-Set Recognition". Tapia Conference, September 2022, Washington DC.
- National Science Foundation (NSF) Scholarship, Tapia Conference 2022.
- Japan Student Services Organization (JASSO) scholarship, Apr 2014 - Mar 2015.